MEALBRIDGE

Surplus Food Rescue System

Implemented DBMS Project Synopsis

Course	Database Management Systems Lab
Program	MCA, First Year, First Semester
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Project idea

Connect food businesses with extra safe food to NGOs that can collect it. A MySQL database and local website keep food offers, claims, and volunteer pickups organised.

Current build

Six MySQL tables, a Python Flask website, claim and pickup validation triggers, a transaction-based claim procedure, and an automatic-expiry event.

1. Introduction and project plan

Cafes, restaurants, and canteens may have extra food at the end of the day. Nearby NGOs may need food for the people they support. When coordination happens only by phone or messages, information can be missed and two organisations may try to collect the same food. MealBridge keeps offers and claims in one shared database.

Aim and objectives

- Store donors, NGOs, food listings, volunteers, and pickups in related tables.

- Show available listings whose expiry time has not passed.

- Let an NGO claim a listing and prevent duplicate claims.

- Assign a volunteer from the claiming NGO to collect the food.

- Track pickup and distribution progress.

- Automatically mark unclaimed, expired listings as EXPIRED.

- Report food offered, claims, and volunteer pickups.

- Practise keys, constraints, joins, triggers, transactions, and scheduled database events.

Technology and plan

MySQL 8 stores the data. Python with Flask serves the website locally. The work creates related tables, adds sample data, implements claim and pickup rules, schedules expiry, builds browser pages, and demonstrates reports.

2. Database design

Six tables hold the main information. Primary keys identify rows, while foreign keys connect related records.

Table	Purpose	Main columns
Donor	Food business offering extra food	donor_id (PK), donor_name, donor_type, phone, area
NGO	Organisation that can collect food	ngo_id (PK), ngo_name, phone, area
FoodListing	Food offer and availability	listing_id (PK), donor_id (FK), food_name, quantity_kg, pickup_area, expiry_time, status
Claim	NGO reservation and progress	claim_id (PK), listing_id (FK, unique), ngo_id (FK), claim_time, claim_status
Volunteer	Volunteer belonging to an NGO	volunteer_id (PK), ngo_id (FK), volunteer_name, phone
Pickup	Volunteer assignment and collection	pickup_id (PK), claim_id (FK, unique), volunteer_id (FK), assigned_time, pickup_time, pickup_status

Relationships and rules

One donor posts many listings; a listing has at most one claim; an NGO makes many claims and has many volunteers; a claim has at most one pickup; a volunteer can perform many pickups. A pickup volunteer must belong to the NGO that claimed the food.

A listing can be claimed only once. Each pickup can be assigned to one volunteer. Quantity must be positive, and foreign keys must point to existing rows.

3. Website, triggers, and automatic expiry

The website registers donors, NGOs, and volunteers; posts food; accepts claims; assigns pickups; records collection and distribution progress; and shows reports.

Claim transaction and trigger

The claim_listing procedure checks availability and expiry, locks the listing, marks it CLAIMED, and inserts the claim in one transaction. A BEFORE INSERT trigger on Claim checks again that the listing is claimed and unexpired. If the operation fails, the procedure rolls back.

Pickup trigger

A BEFORE INSERT trigger on Pickup checks that the claim belongs to the volunteer's NGO and is ready for pickup. This protects the rule even if a pickup row is inserted outside the website.

Automatic expiry event

The expire_unclaimed_listings MySQL event runs every minute. It changes AVAILABLE listings to EXPIRED once the expiry time passes. A trigger cannot run just because time passes, so MySQL Event Scheduler handles this timed task.

Reports and other DBMS concepts

Reports show kilograms offered by each donor, claims by NGO, and assigned/completed pickups by volunteer. Primary keys identify rows; foreign keys link tables; constraints protect values; JOIN combines related records; SUM and COUNT calculate report totals; transactions save or roll back related changes.

4. Scope choices and conclusion

The project is kept small enough for first-semester MCA students to understand and present. These parts of the original concept are still not included:

Accounts and administrator verification: Donors and NGOs can be registered directly; there are no user logins or admin approval screens.

Separate Distribution table: There is no beneficiary count or distribution record. Distribution progress is stored as a Claim status.

Advanced impact reports: The site does not calculate rescued-food totals, average response time, or NGO rankings.

Mobile and location features: The local website does not provide maps, nearby matching, or notifications.

Conclusion

MealBridge demonstrates how a relational database, triggers, a scheduled event, and a small website can organise surplus food listings, NGO claims, and volunteer pickups. Its six-table design gives students practical examples of DBMS concepts.

Possible future improvements

Administrator verification, beneficiary distribution records, mobile support, maps, and response-time reports can be added in a later version if required.